Housing Market Shocks and the Persistent Decline in U.S. Fertility

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November 6, 2026

Opinion

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Births and immigration are slowing at the same time. The U.S. isn't ready for what that means.

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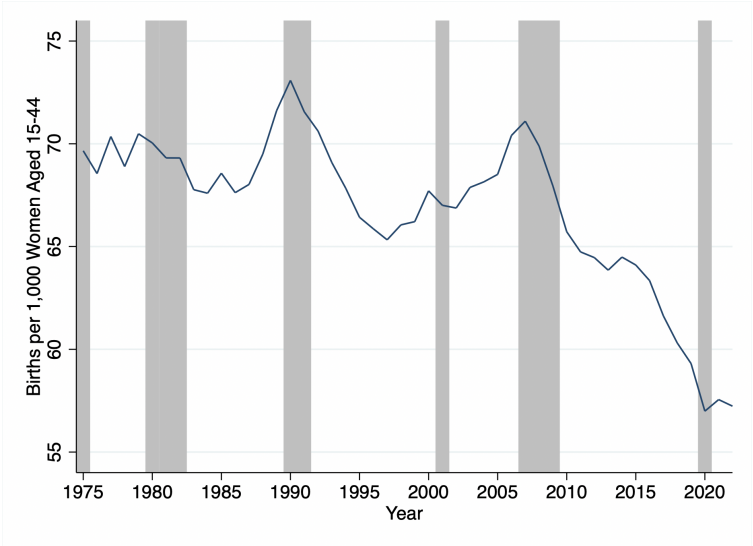
Hungary's Orban launches tax exemption for mothers, cap on housing loan rates

Reuters, February 22, 2025

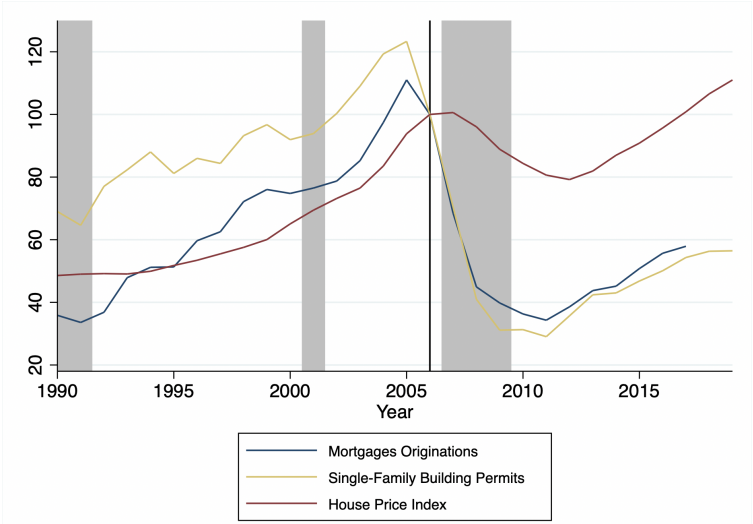
Seoul boosts fertility treatment support by 77% to tackle low birthrate

The Korea Times, March 11, 2024

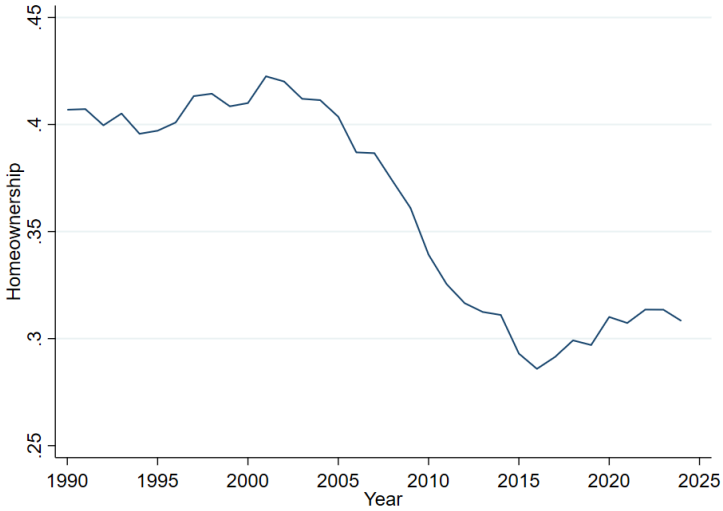
Fertility declined after the Great Recession and never recovered



Contemporaneous housing market bust



Decline in Homeownership among Women Ages 15–44



Did housing bust contribute to the persistent decline in fertility?

- Exploit variation across MSAs in exposure to the housing bust
  - Decline in mortgage originations per capita 2006–2011
  - More exposed MSAs experienced larger housing boom-bust dynamics
  - Alternative measures of housing-bust exposure
- Compare changes in fertility over time across more and less exposed MSAs using an event-study framework
- Examine housing and household outcomes to explore potential channels

Housing market and fertility decisions are closely linked

- Housing market conditions can affect fertility through multiple channels:
  - Dettling and Kearney (2014) find that the house price increases during the housing boom led to increased fertility among homeowners but reduced fertility among non-homeowners
  - Japaridze and Sayour (2024) show that renters delay childbearing relative to homeowners, with delays being greater in less affordable housing markets
  - Dettling and Kearney (2026) show that the introduction of federally backed mortgages led to earlier homeownership and increased fertility
- Difficulties accessing suitable housing may delay partnership formation and childbearing

Housing bust and fertility

- Falling house prices:
  - Existing homeowners: Reduced housing wealth/equity $\Rightarrow$ Fertility $\downarrow$
  - Underwater homeowners: Constrained mobility, difficult to adjust housing to expand families $\Rightarrow$ Fertility $\downarrow$
  - Prospective buyers: Housing more affordable $\Rightarrow$ Fertility $\uparrow$
- Tighter mortgage credit:
  - Prospective buyers: Harder to purchase homes and establish independent households $\Rightarrow$ Fertility $\downarrow$
- Reduced housing construction:
  - Fewer new homes lead to more difficulty in finding suitable housing $\Rightarrow$ Fertility $\downarrow$

Preview of Results

- More exposed MSAs experienced larger declines in fertility
  - Effects persist through the end of the sample in 2022
  - Back-of-the-envelope calculation suggests the housing bust could account for 38% of the fertility decline by 2011, and 26% by 2022
- Results are robust to alternative measures of housing-market exposure, and are not explained by local labor-market deterioration
- Fertility declines occur for both first and higher-order births and are concentrated among women under age 35
- More exposed MSAs experienced persistent changes in housing and household outcomes
  - Increase in co-residence with parents
  - Decline in partnership and homeownership
  - Decline in residential mobility

Related Literature

- Housing markets and fertility: Lovenheim and Mumford (2013); Dettling and Kearney (2014, 2026); Cumming and Dettling (2023); Japaridze and Sayour (2024); Kim, Kim, and Park (2024)
- Economic conditions and fertility: Lindo (2010); Kearney and Wilson (2018); Kearney, Levine, and Pardue (2022)
- Long-run effects of the Great Recession: Yagan (2019); Hershbein and Kahn (2018); Rothstein (2023)

- **This paper:** Examines whether housing-market disruptions contributed to the persistent post-recession fertility decline, beyond the immediate effects of housing and labor-market shocks

Outline

1. Data
2. Empirical Method
   - Defining Housing Market Exposure
   - Characterizing Housing Market Exposure
   - Event-Study Specification
3. Main Results
4. Labor Market Conditions
5. Housing and Household Responses

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  - Census, American Community Survey (ACS)
- Unit of analysis: Metropolitan Statistical Areas (MSAs)
  - Approximate local housing and labor markets
  - Main analysis includes 203 unique MSAs
  - Results robust to county-level analysis (not shown today)

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1a. *Increase* in home-purchase mortgage originations per 1,000 residents between 2000 and 2006
   - Highly correlated with the preferred measure, $\rho = 0.9$
   - Addresses timing of measurement window rather than providing independent variation

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- These measures reflect equilibrium housing-market outcomes, rather than exogenous shocks

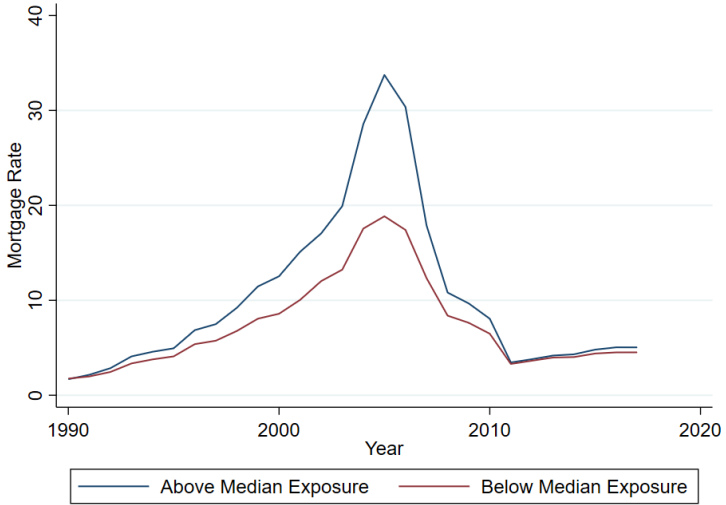
Measures capture overlapping but distinct dimensions

	Δ Mortgage Origination Rate	Δ Permit Rate	Subprime Rate
Δ Permit Rate	0.7691		
Subprime Rate	0.3657	0.1658	
Δ HPI	0.4675	0.4539	0.1560

More exposed MSAs had greater housing-market activity pre-crisis

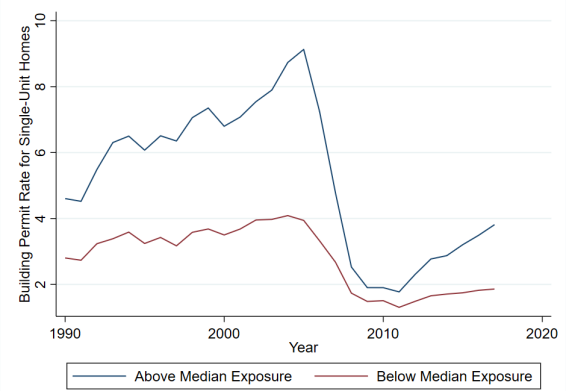
VARIABLES	Δ Mortgage Originations
Employment Rate	0.0107
	(0.0477)
Construction Share of Employment	182.4***
	(50.48)
Personal Income	-0.0507**
	(0.0240)
HPI	0.000752
	(0.00390)
Fertility Rate	0.0536
	(0.0356)
ln(MSA Population)	2.344***
	(0.550)
Mortgage Originations	1.499***
	(0.129)
Single-Family Building Permits	0.324***
	(0.0940)
N	203

More exposed MSAs experienced larger mortgage booms and busts

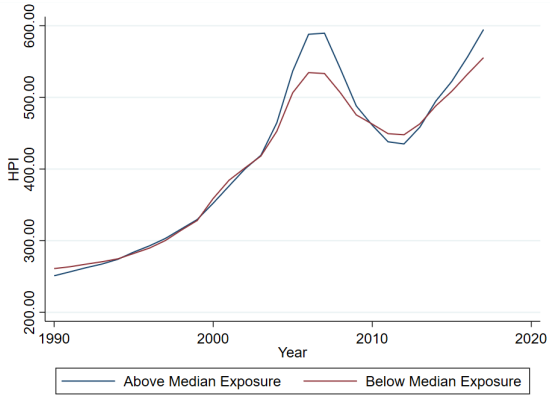


More exposed MSAs experienced broader housing-market disruptions

(a) Single-Family Building Permits



(b) HPI



Event-Study Specification

$$Y_{mgt} = \sum_{k \neq -1} \beta_k \mathbb{1}\{k = t - 2007\} \, Exposure_m + X_{m,2000} \times \gamma_t + \delta_m + \delta_t + \delta_g + \varepsilon_{mgt}, \qquad (1)$$

where

- Y_{mgt}: fertility rate for race-by-age-group cell g in MSA m in year t
- $Exposure_m$ is a measure of MSA m's exposure to the housing bust
- $X_{m,2000}$ is a vector of baseline MSA characteristics measured in 2000, including the mortgage origination rate, single-family permit rate, housing price index, personal income, employment rate, and construction employment share
- δ_m, δ_t, and δ_g are MSA, year, and race-by-age-group FEs
- Regressions are weighted by race-by-age-group female population, and standard errors are clustered at the MSA level

Identification

- Identifying assumption
  - Absent the housing bust, fertility trends would have evolved similarly across MSAs with different levels of exposure, conditional on baseline characteristics
- Threats to identification
  - MSAs more exposed to the housing bust would have experienced differential fertility trends even in the absence of the bust
  - Other contemporaneous shocks correlated with housing-market exposure independently affect fertility
- To assess validity
  - Test sensitivity to baseline MSA characteristics and time-varying controls
  - Examine differential pre-trends in fertility
  - Account for differences in local labor-market shocks

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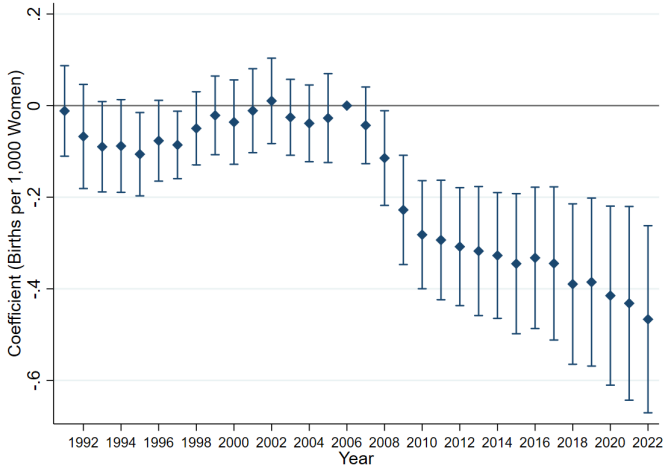
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More exposed MSAs experienced larger fertility declines



Alternative Specifications

How large is the fertility decline?

- Estimated fertility effect
  - By 2011: -0.29 births per 1,000 women per unit of exposure
  - By 2022: -0.47 births per 1,000 women per unit of exposure

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- Actual fertility decline relative to 2006
  - 5.83 by 2011 and 13.60 by 2022

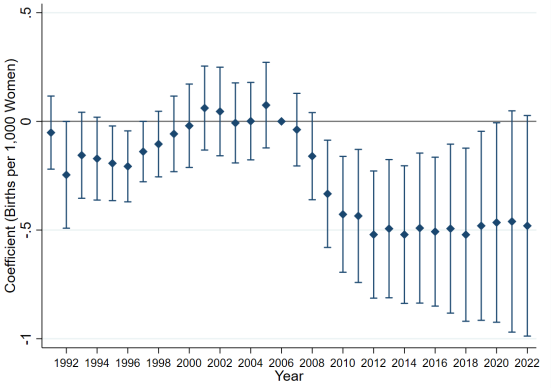
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  - 5.83 by 2011 and 13.60 by 2022
- Under this counterfactual, the additional contraction could account for 38.1% of the fertility decline by 2011 and 25.9% by 2022

Details

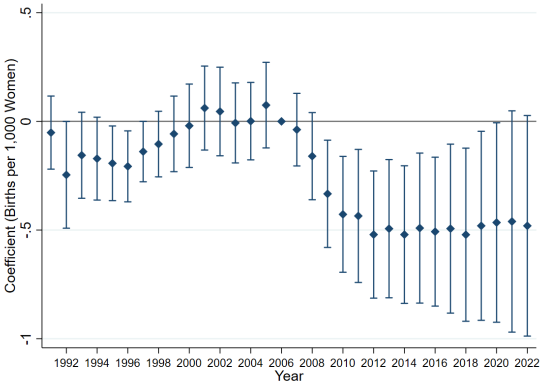
Alternative Housing-Market Exposures

(a) Decline in Building Permits

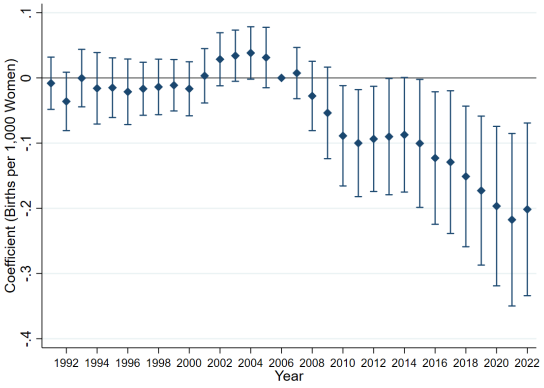


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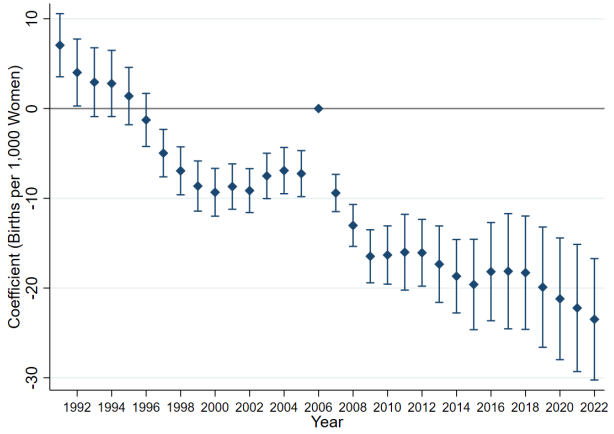


(b) Baseline Subprime Rates



Mortgage Originations Increase

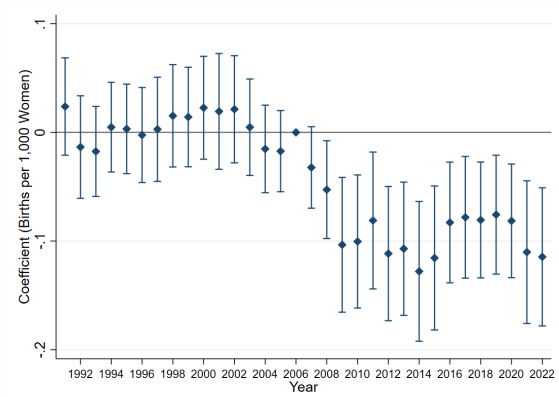
Alternative Housing-Market Exposures: Housing Price Decline



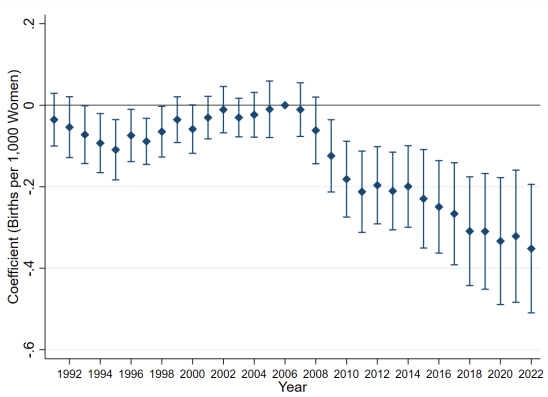
- Pre-trends suggest that HPI exposure captures broader housing-market boom-bust dynamics correlated with fertility, but does not isolate the fertility decline following the Great Recession

Declines in Both First and Higher-Order Births

(a) First Births



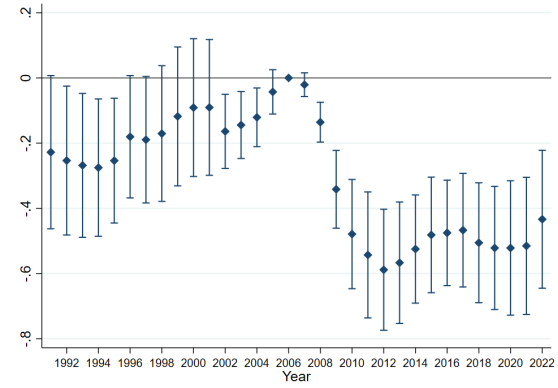
(b) Higher-Order Births



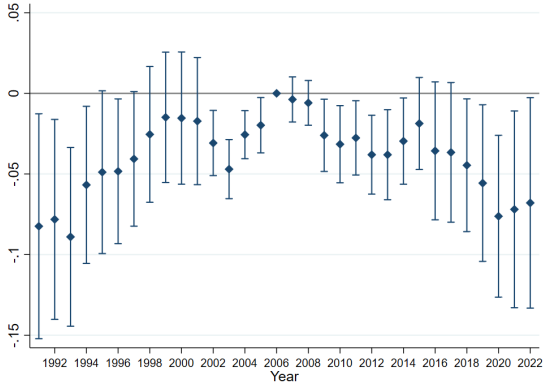
- Mean fertility rates in 2006: 25.14 for first births and 38.95 for higher-order births (per 1,000 women)

Effect Concentrated Among Women under 35

(a) Ages 20–34



(b) Ages 35–44



By 5-year Age Bins

Takeaway: MSAs more exposed to the housing bust experienced larger and persistent declines in fertility, affecting both first and higher-order births, particularly among women under 35.

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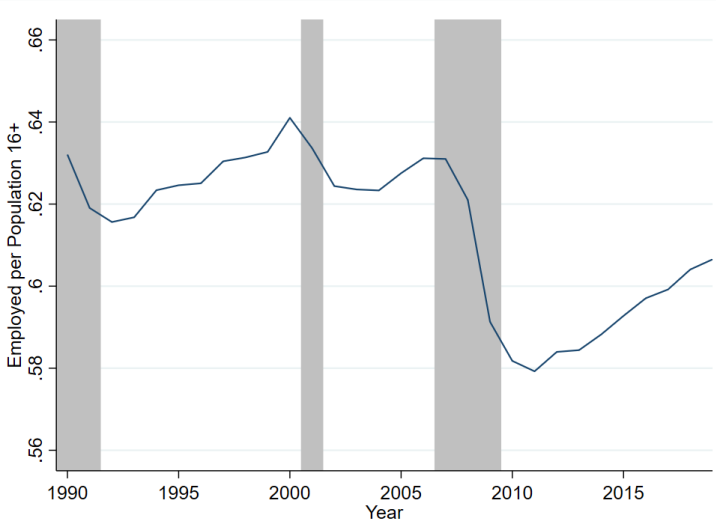
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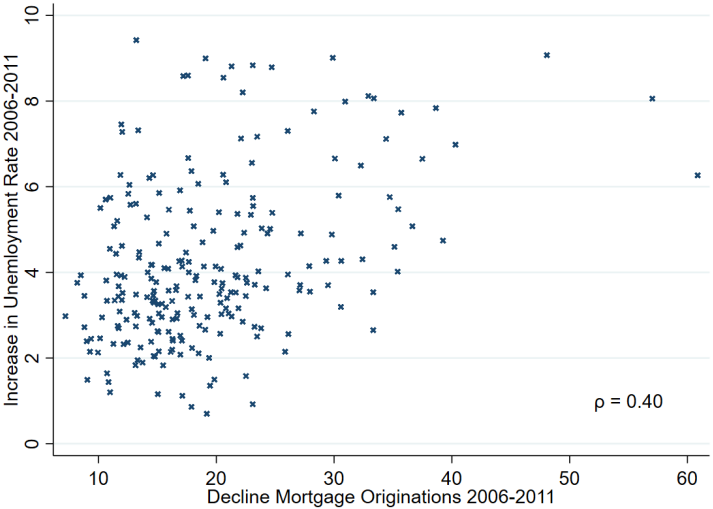
Slow recovery in employment post-Great Recession



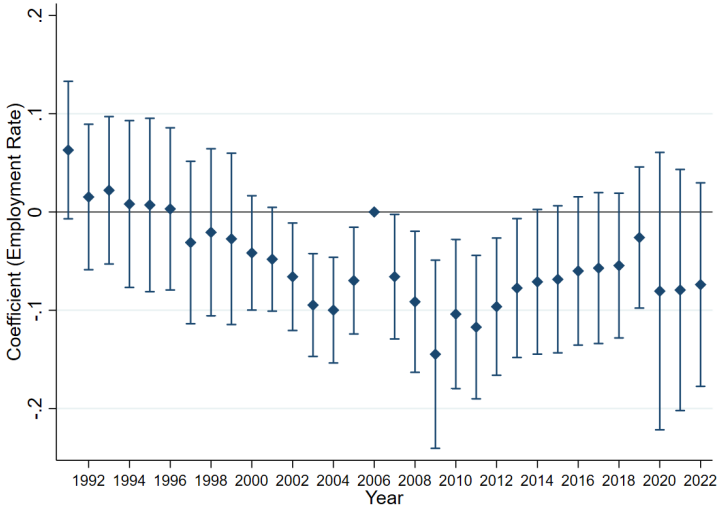
Can the Fertility Decline be Explained by Labor Market Conditions?

- The Great Recession combined a severe housing-market contraction with a sharp deterioration in local labor markets
- Local exposure to the Great Recession had persistent effects on employment (Yagan 2019)
- Labor-market conditions are a known determinant of fertility (Lindo 2010; Kearney and Wilson 2018)
- Are the fertility effects driven by broader local labor-market shocks rather than housing-market shocks?

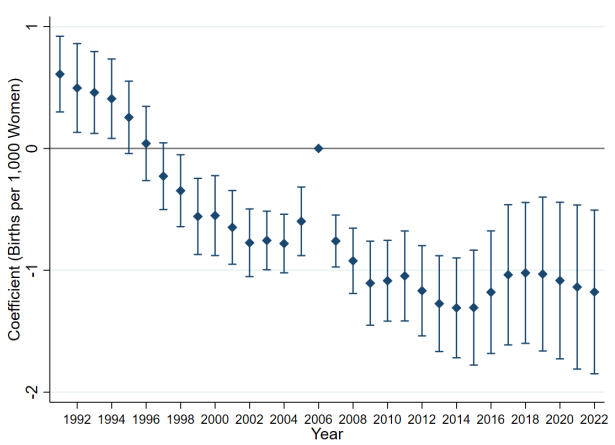
Housing and Labor-Market Exposures Not Highly Correlated



Housing-market exposure not associated with persistent employment declines

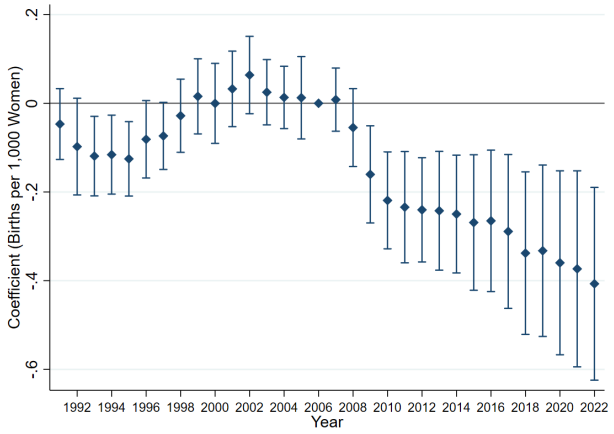


Fertility Event Study using Labor-Market Exposure



- MSAs with larger unemployment shocks exhibit substantial differential pre-trends, suggesting broader boom-bust patterns rather than a distinct post-recession fertility decline

Controlling for Labor-Market Shocks in the Main Specification



- The estimated housing-market effect remains substantial and statistically significant after controlling for labor-market exposure (-0.276 vs. -0.225)

Takeaway: The larger fertility declines in MSAs experiencing more severe housing busts are not explained solely by contemporaneous labor-market shocks.

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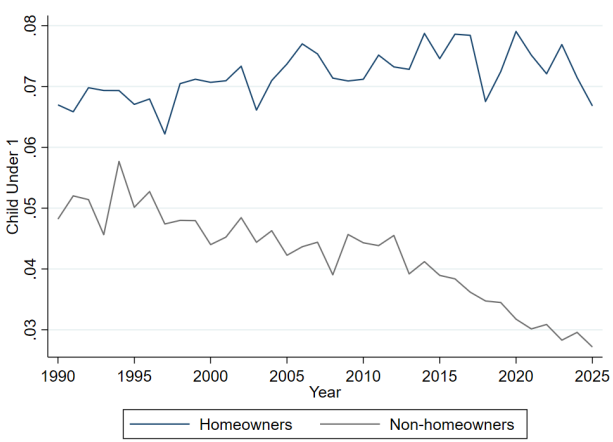
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Fertility Declines *Within* Non-Homeowners



- Fertility remains relatively stable among homeowners, but declines *among* non-homeowners
- Changes in homeownership alone cannot account for the full national fertility decline

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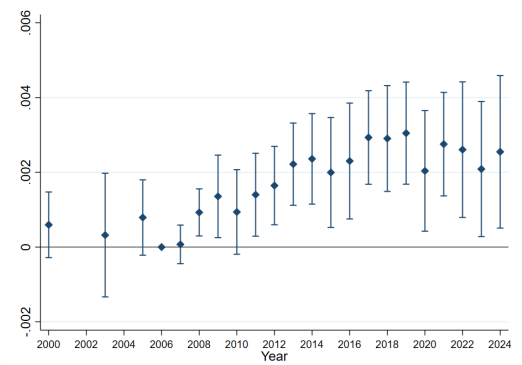
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  - (iv) Moves into owner-occupied housing: Moved within 1 year & Owns home

Housing-Market Shocks and Household Formation

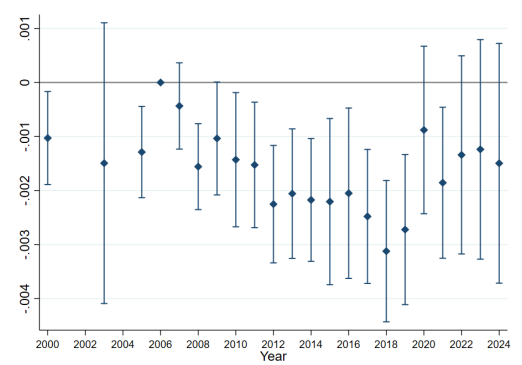
- Housing-market disruptions may affect fertility by delaying residential independence and household formation
- Examine household and housing responses using Census and ACS, focusing on young adults ages 20–34
  - (i) Parental coresidence: Whether the individual lives with at least one parent
  - (ii) Partnered household formation: Whether individual is householder with a spouse or unmarried partner, or a spouse or unmarried partner of a householder
  - (iii) Homeownership: Is householder or spouse, and owns residence
  - (iv) Moves into owner-occupied housing: Moved within 1 year & Owns home
  - (v) Residential mobility: Moved within 1 year

Housing-Market Shocks Delay Household Formation

(a) Parent with Household

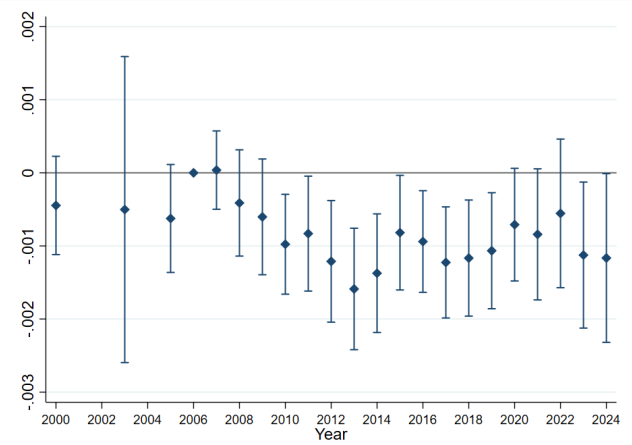


(b) Partnered Household



- The additional contraction in mortgage originations below pre-boom levels is associated with a 0.87pp increase in parental coresidence, equivalent to 14.7% of the observed pre-post increase
- A 0.74pp decline in partnered household formation, equivalent to 13.8% of the observed pre-post decline

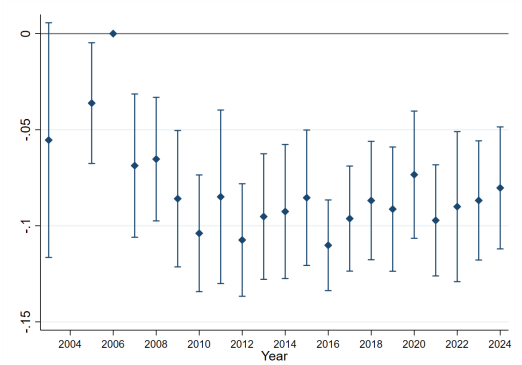
Reduced Homeownership



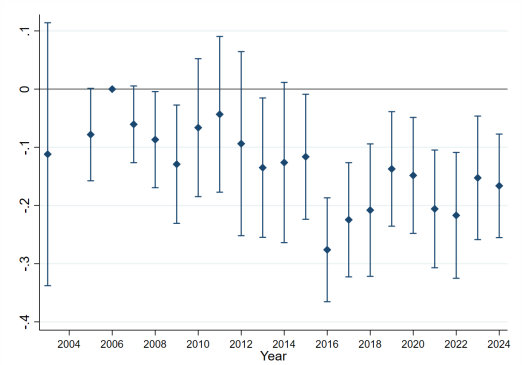
- The additional contraction in mortgage originations below pre-boom levels is associated with a 0.80pp decline in homeownership, equivalent to 10.0% of the observed pre-post decline

And Reduced Residential Mobility

(a) Moved Within 1 Year & Owns Home



(b) Moved Within 1 Year



- The additional contraction in mortgage originations below pre-boom levels is associated with a 0.49pp decline in moves into owner-occupied residences, equivalent to 29.5% of the observed pre-post decline
- A 0.50pp decline in total residential mobility, equivalent to 13.3% of the observed pre-post decline

Takeaway: MSAs experiencing larger housing busts also experienced delayed household formation, reduced homeownership, and reduced residential mobility, particularly moves into owner-occupied housing.

Discussion

- Cannot separately identify the effects of mortgage credit, residential construction, and house prices
  - Results suggest that house price changes alone are unlikely to fully explain the fertility decline
  - HPI-based exposure exhibits substantial differential pre-trends, and fertility effects persist as house prices recover

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  - Results suggest that house price changes alone are unlikely to fully explain the fertility decline
  - HPI-based exposure exhibits substantial differential pre-trends, and fertility effects persist as house prices recover
- Household responses are consistent with broader disruptions to housing access and housing transitions
  - Decline in homeownership *and* decline in fertility among non-owners
  - Potential roles for expectations about future homeownership, access to suitable housing, and delayed household and partnership formation

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- Household responses are consistent with broader disruptions to housing access and housing transitions
  - Decline in homeownership *and* decline in fertility among non-owners
  - Potential roles for expectations about future homeownership, access to suitable housing, and delayed household and partnership formation
- Fertility responses are asymmetric and persistent
  - MSAs more exposed to the housing bust also experienced larger pre-crisis housing booms, but without corresponding differential fertility increases
  - Effects persist through 2022, even among women who were under 20 during the housing bust

Conclusion

- Housing-market disruptions contributed to the persistent decline in U.S. fertility following the Great Recession
  - Declines in both first and higher-order births
  - The additional mortgage-market contraction in below pre-boom levels could account for 38% of the fertility decline by 2011 and 26% of the decline by 2022

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- Fertility declines extend beyond the immediate recession
  - Results remain after accounting for local labor-market shocks
  - Differential fertility declines persist through 2022, even among women who were children when the bust began
- Housing-market disruptions can have lasting demographic consequences
  - Housing access and household formation may shape fertility decisions well beyond the initial economic shock

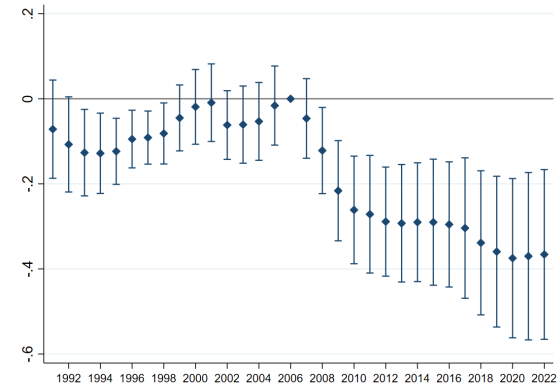
Research Agenda

Labor and health economics

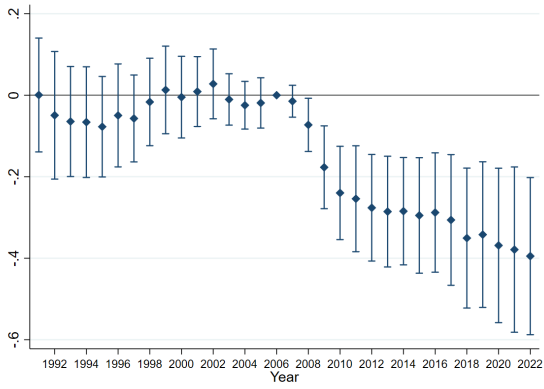
- **Fertility and family formation**
  - Housing-market shocks and fertility
  - Access to contraception and reproductive decisions
  - Sex education and fertility
- **Health policy and consumer behavior**
  - Tobacco regulation
  - Health insurance coverage and health care utilization

Alternative Specifications

(a) Without Baseline Controls



(b) With Time-Varying Controls



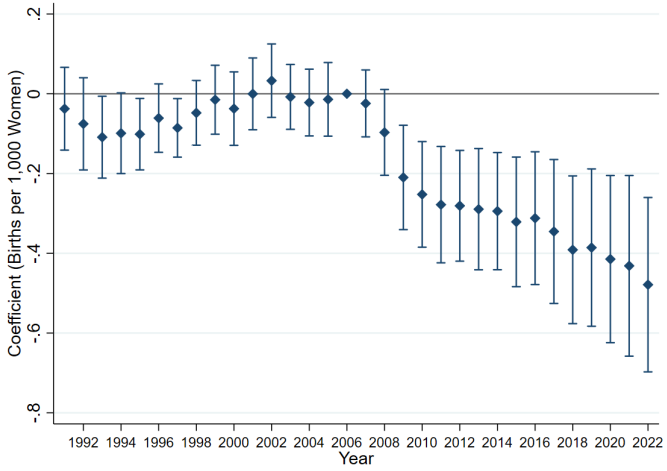
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How large is the fertility decline?

	2000	2006	2011	2022
Mortgage originations	11.01	26.21	3.46	
Relative to 2000			−7.55	
Fertility rate		69.77	63.94	56.17
Relative to 2006			−5.83	−13.60
Event-study coefficient			−0.2934	−0.4664
Implied change in fertility			−2.22	−3.52
Share of observed decline			38.1%	25.9%

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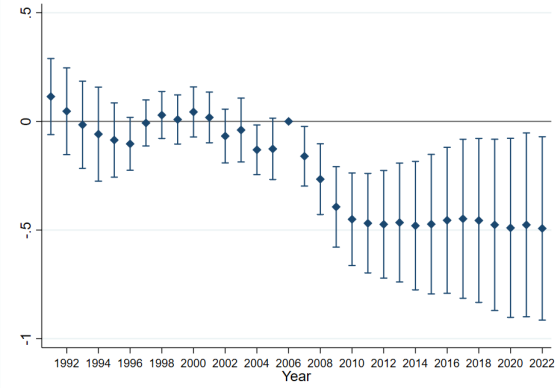
Alternative Specifications: Mortgage Originations Increase



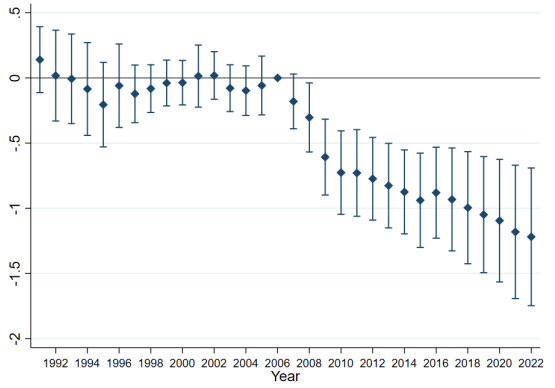
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Effect by 5-year age bins

(a) Ages 15–19



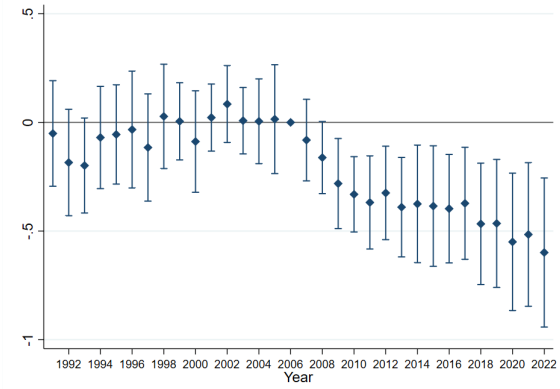
(b) Ages 20–24



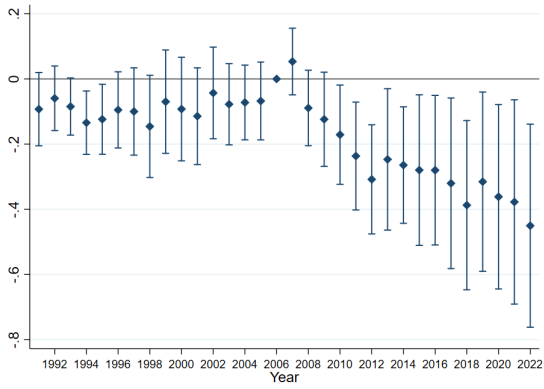
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Effect by 5-year age bins

(c) Ages 25–29



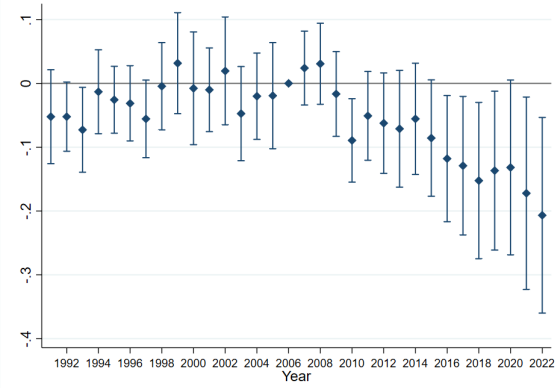
(d) Ages 30–34



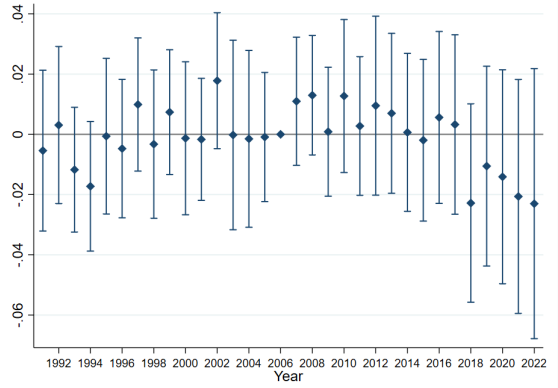
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Effect by 5-year age bins

(e) Ages 35–39



(f) Ages 40–44



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